

[Aim: 100/100 in Maths]

आरंभ

① BPT

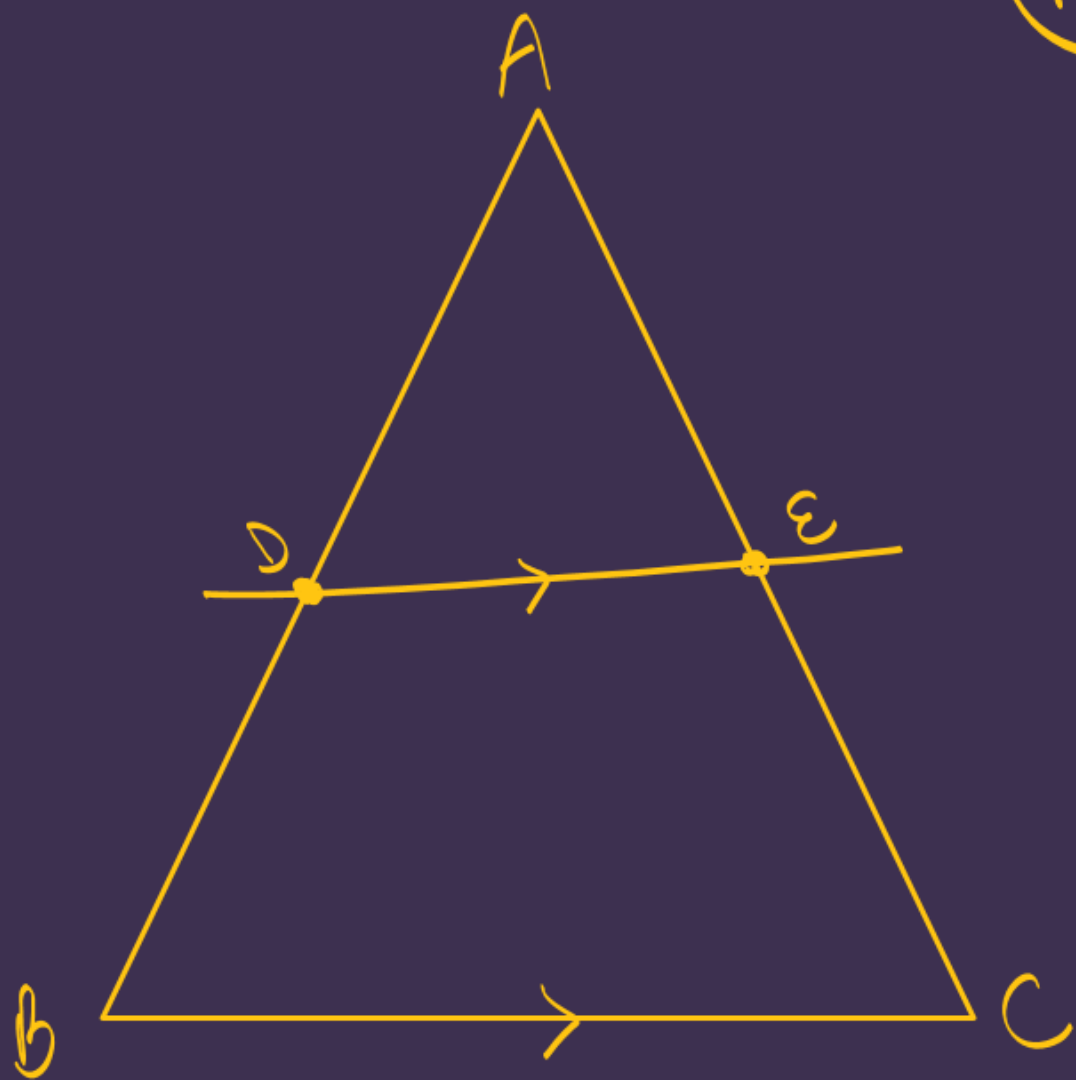
② Similarity

TRIANGLES

Lecture #1

#BASIC PROPORTIONALITY THEOREM: (BPT)

If a line is drawn parallel to one side of a triangle intersecting the other two sides in distinct points, then the other two sides are divided in the same ratio.



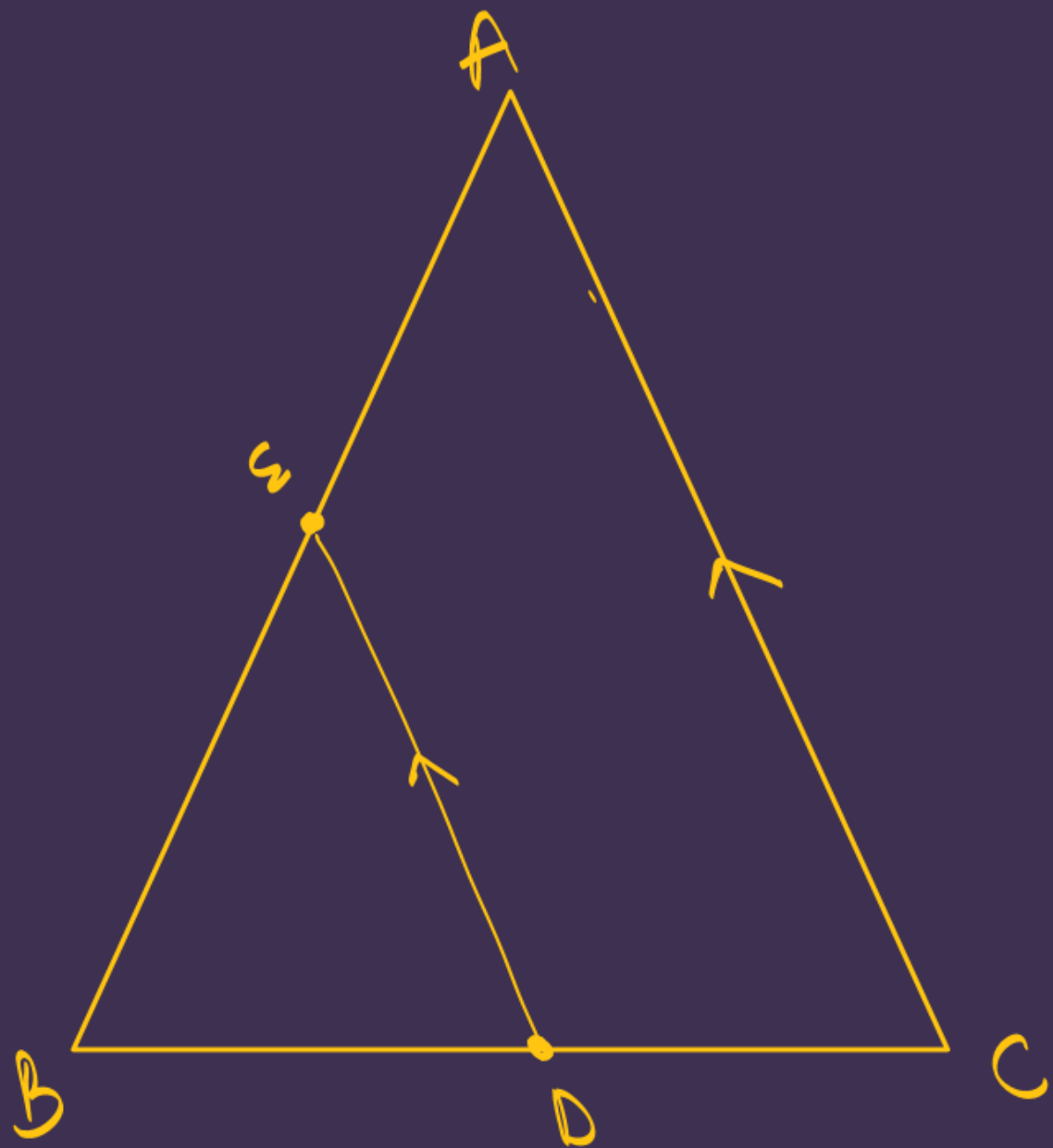
pf $DE \parallel BC$

$$\frac{AD}{DB} = \frac{AE}{EC}$$

$$\frac{AD}{AB} = \frac{AE}{AC}$$

$$\frac{DB}{AB} = \frac{EC}{AC}$$

by
BPT



if $\boxed{ED \parallel AC}$

→

$$\frac{BD}{DC} = \frac{BE}{EA} \quad \checkmark$$

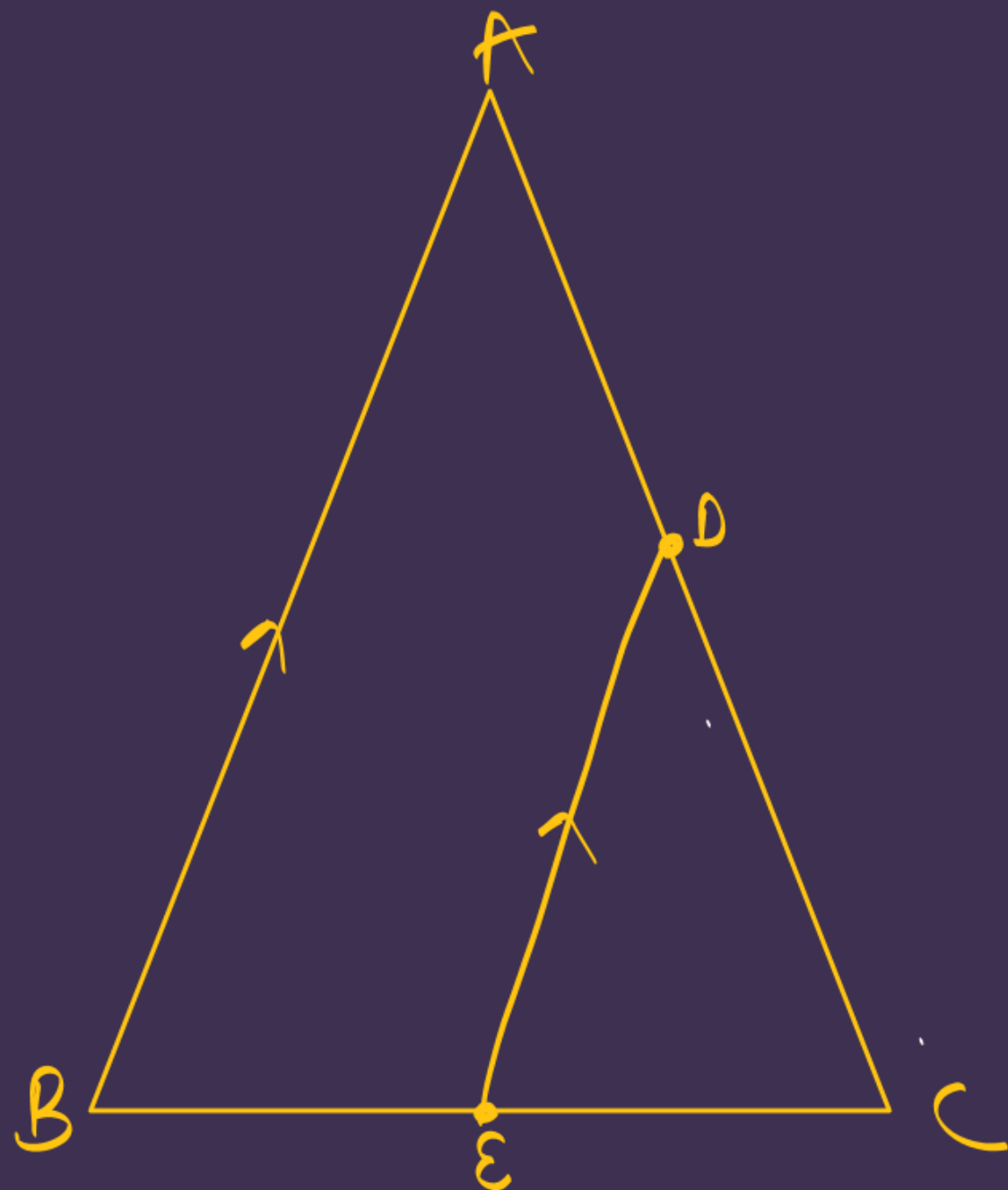
→

$$\frac{BD}{BC} = \frac{BE}{BA} \quad \checkmark$$

→

$$\frac{DC}{BC} = \frac{EA}{BA} \quad \checkmark$$

by
BPT



if $DE \parallel AB$



$$\frac{CE}{EB} = \frac{CD}{AD} \checkmark$$

$$\frac{CE}{CB} = \frac{CD}{CA} \checkmark$$

$$\frac{EB}{CB} = \frac{DA}{CA} \checkmark$$

by BPT.

#LP: In Fig., if $ST \parallel QR$. Find **PS**.

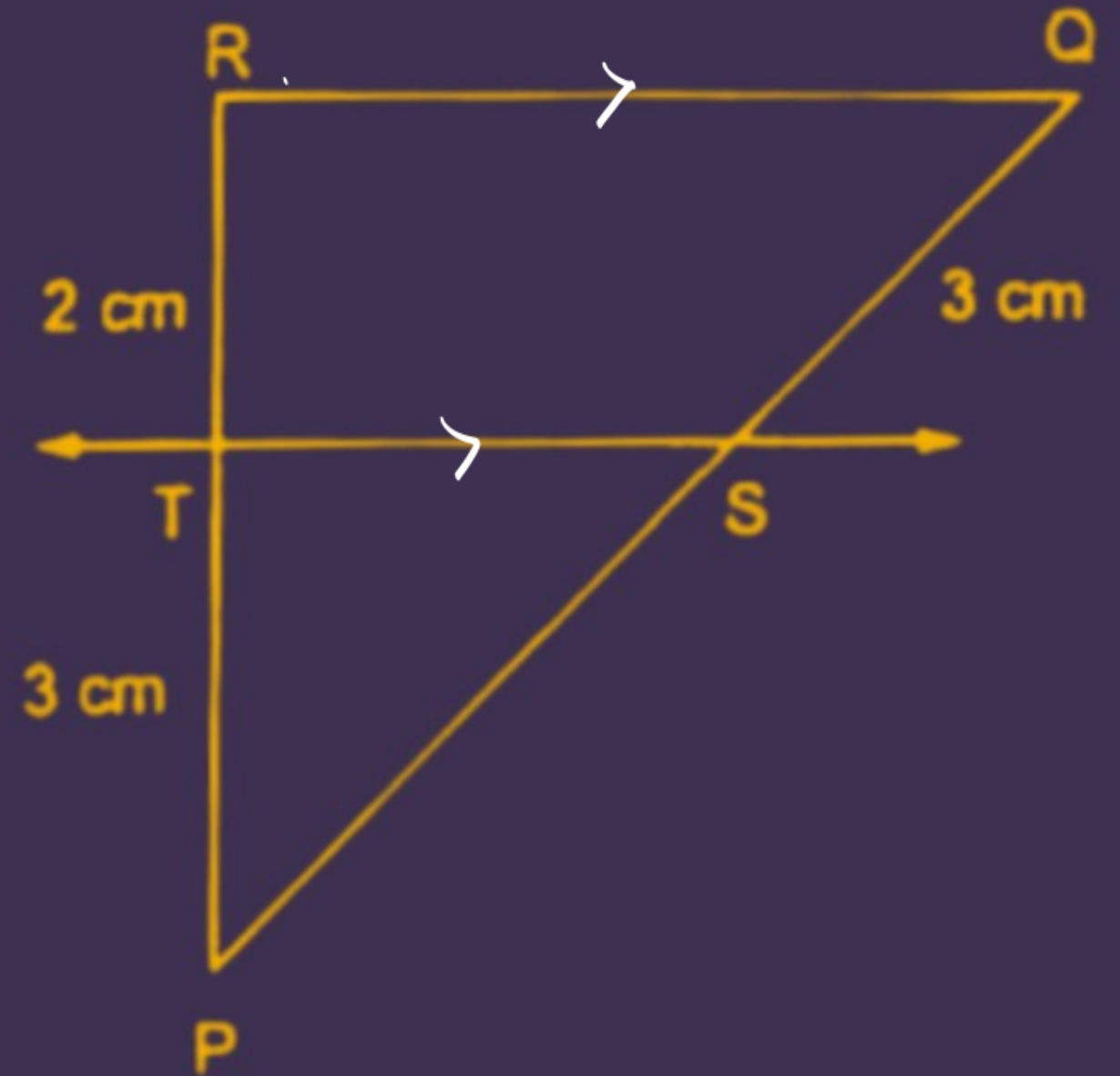
Since $ST \parallel QR$

$\therefore$ by BPT $\rightarrow \frac{PT}{TR} = \frac{PS}{SQ}$

$$\Rightarrow \frac{2}{3} = \frac{PS}{3}$$

$$\Rightarrow \frac{9}{3} = PS$$

$$\boxed{PS = 4.5 \text{ cm}}$$



#LP: **DE || BC**. Find **AD : BD**

A. 1:1

~~B. 2:1~~

C. 3:2

D. 4:3

BPT ✓

$$\frac{AD}{DB} = \frac{AE}{EC}$$

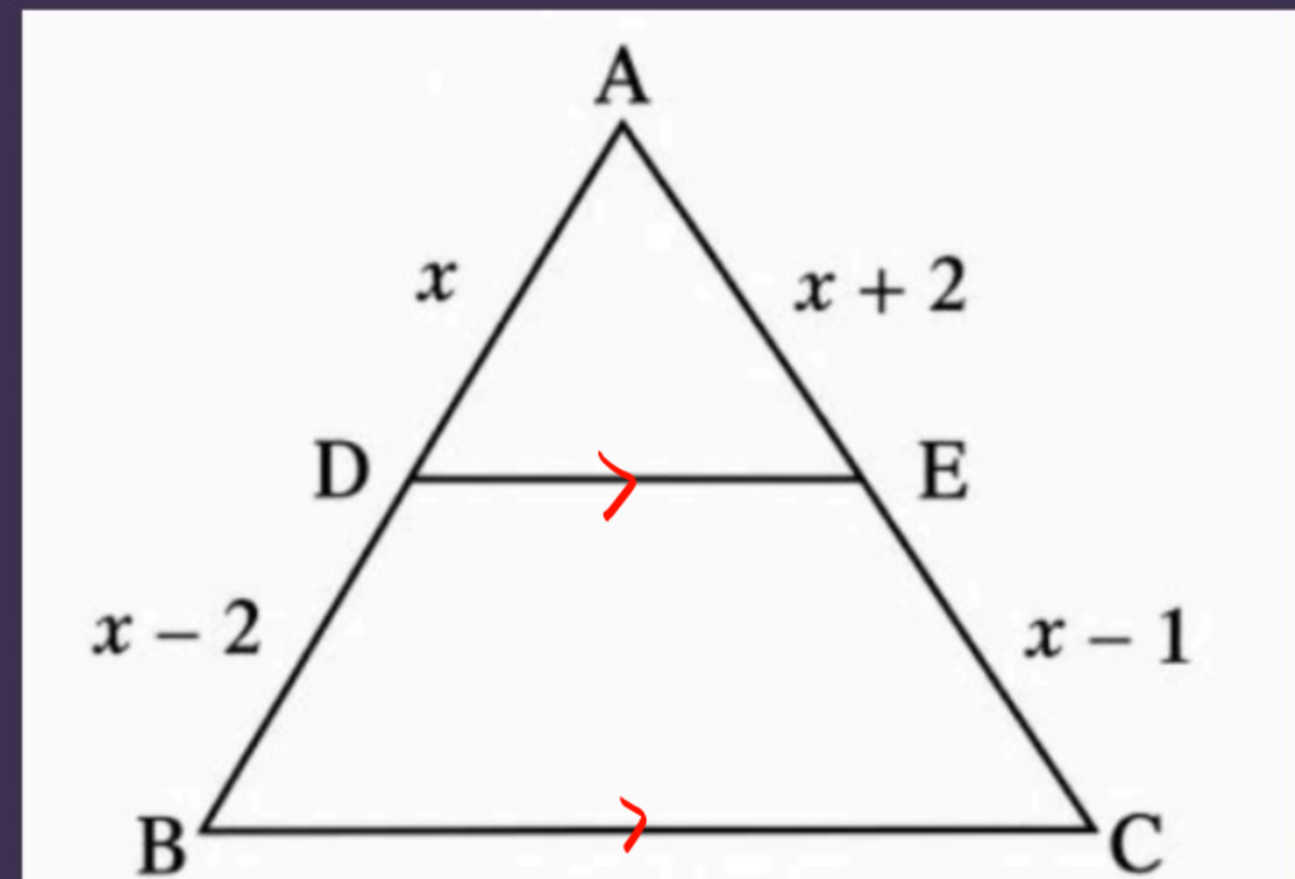
$$\Rightarrow \frac{x}{x-2} = \frac{x+2}{x-1}$$

$$\Rightarrow x(x-1) = (x+2)(x-2)$$

$$\Rightarrow x^2 - x = x^2 - 2^2$$

$$x = 4$$

$$x = 4$$



$$\frac{AD}{DB} = \frac{x}{x-2} \Rightarrow \frac{4}{4-2}$$

$$\Rightarrow \frac{4}{2} = 2 \Rightarrow 2:1$$

#LP: In Triangle PQR. If $XY \parallel QR$, $PQ/XQ = 7/3$ and $PR = 6.3\text{cm}$, Find YR.

A. 2.1 cm

✓ B. 2.7 cm

C. 3.6 cm

D. 4.2 cm

$$\frac{PQ}{XQ} = \frac{7}{3}$$

$$\frac{XQ}{PQ} = \frac{3}{7}$$

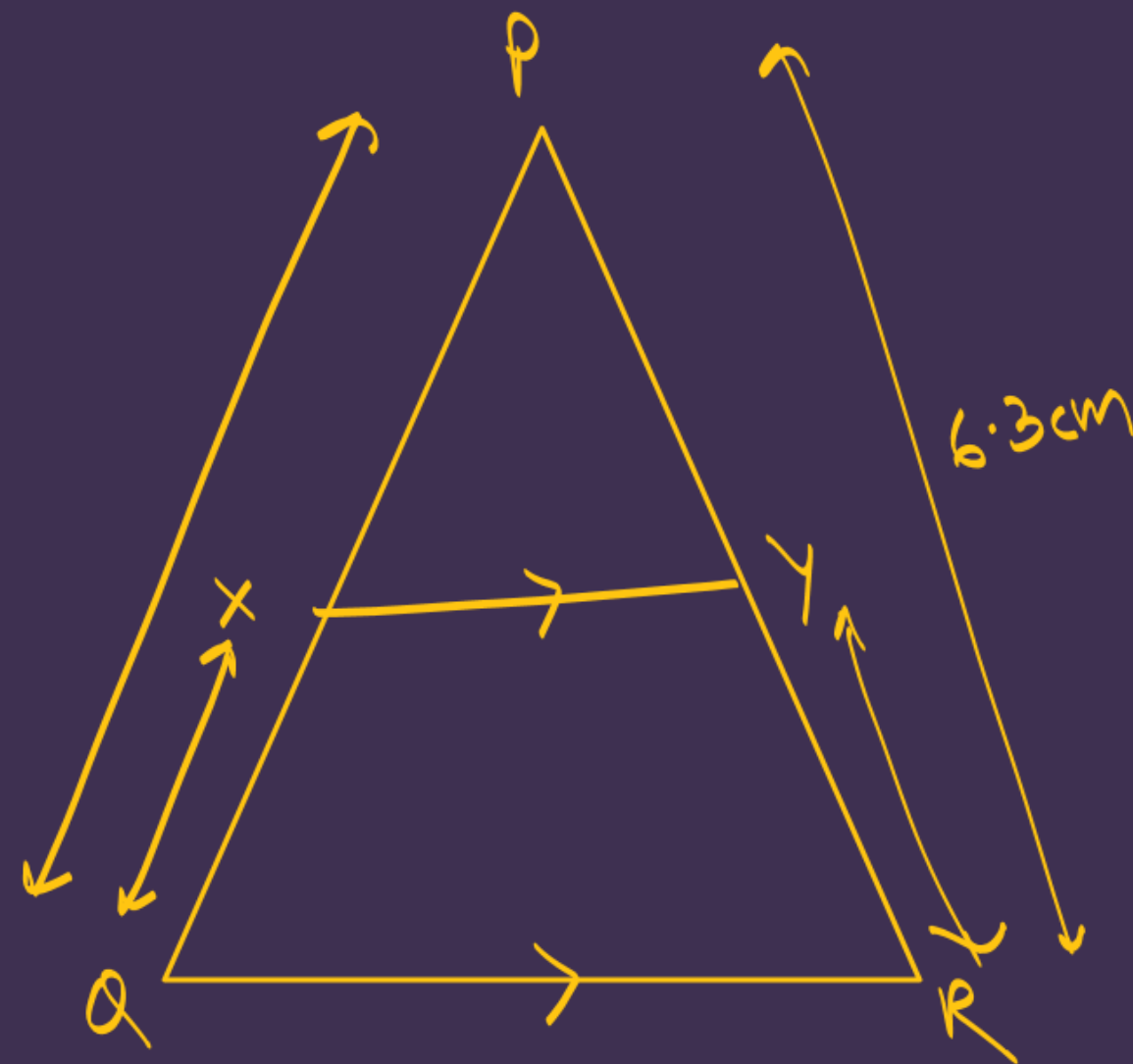
BPT $\Rightarrow$

$$\frac{XQ}{PQ} = \frac{YR}{PR}$$

$$\frac{3}{7} = \frac{YR}{6.3}$$

$$\frac{3}{7} \times 6.3 = YR$$

$$YR = \frac{27}{10} \Rightarrow \underline{\underline{2.7\text{cm}}}$$



#LP: In the fig, $EF \parallel AB \parallel DC$. Prove that $\frac{AE}{ED} = \frac{BF}{FC}$.

① In $\triangle ADC$
 Since $EF \parallel DC$
 $\therefore EP \parallel DC$

by BPT $\rightarrow \frac{AE}{ED} = \frac{AP}{PC}$ (I)

Now In $\triangle CAB$

given $\rightarrow EF \parallel AB$
 $\therefore PF \parallel AB$

by BPT $\Rightarrow \frac{CP}{PA} = \frac{CF}{FB}$

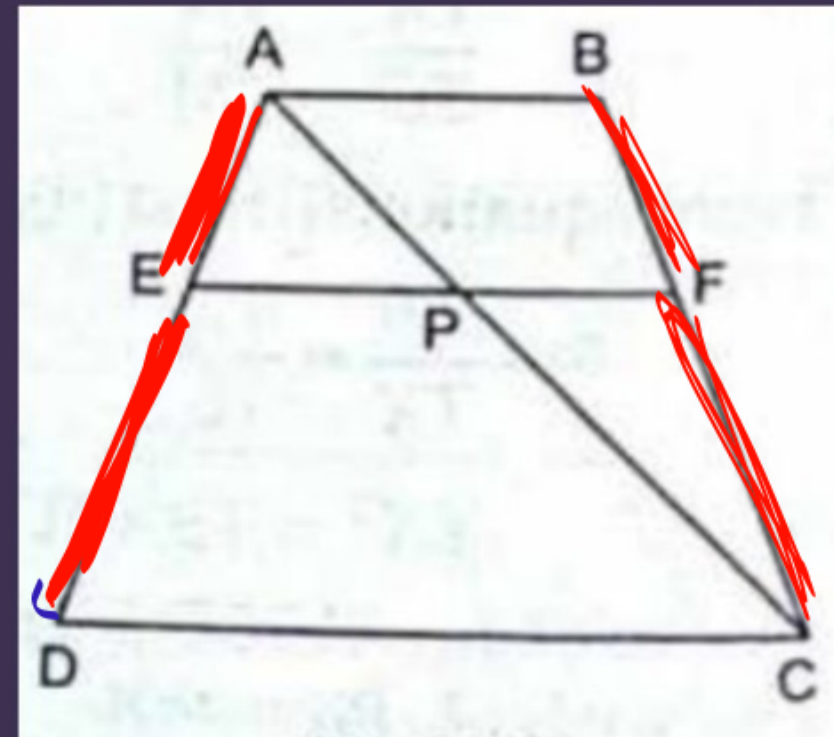
reciprocal

$\frac{PA}{CP} = \frac{FB}{CF}$ (II)

from (I) & (II)

$\frac{AE}{ED} = \frac{FB}{CF}$

HP



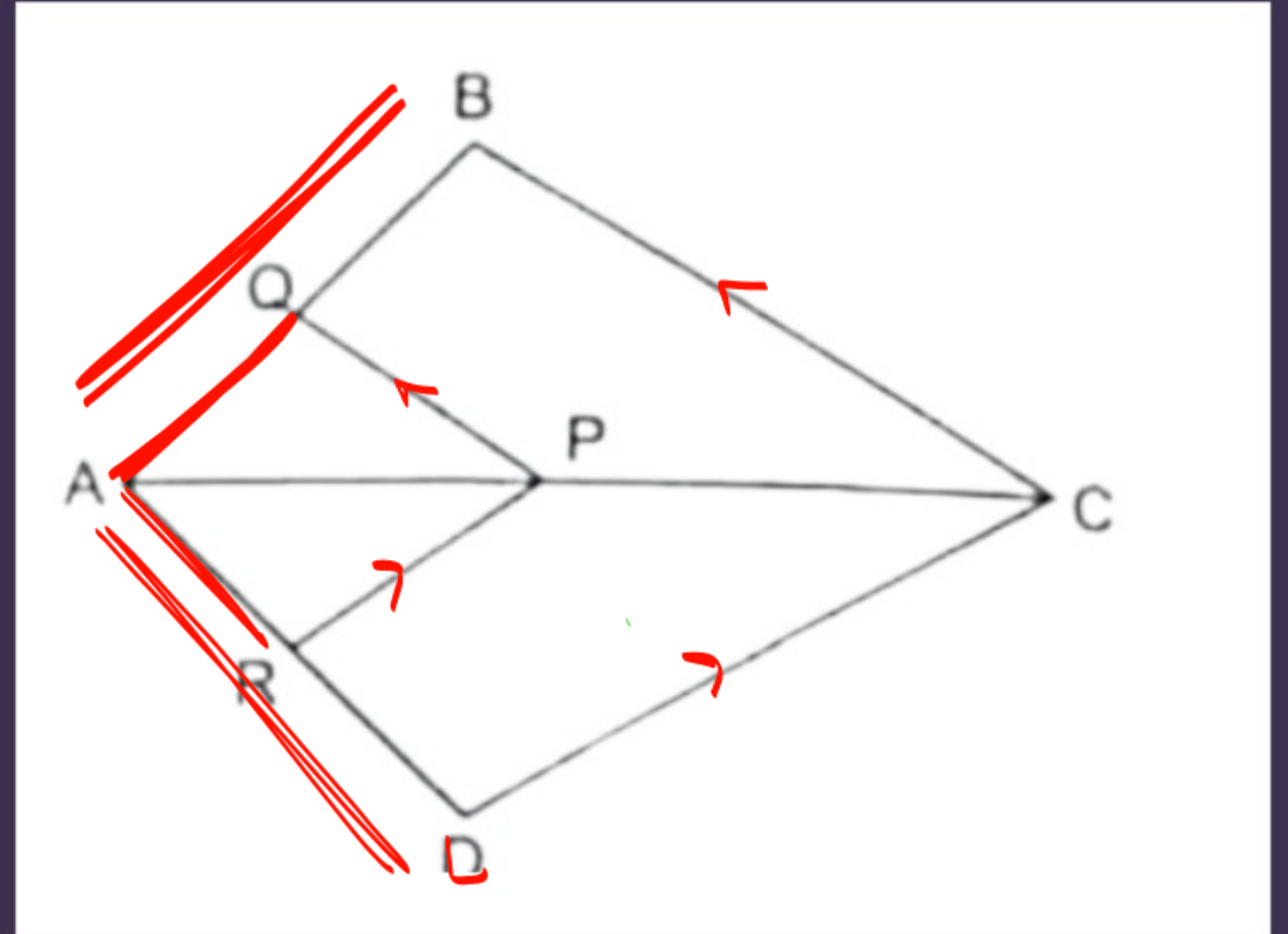
#LP: In the fig, if $PQ \parallel BC$ and $PR \parallel CD$. Prove that

i. $\frac{AR}{AD} = \frac{AQ}{AB}$ ii. $\frac{AP}{AC} = \frac{AR}{AD}$

[CBSE 2010]

$$\frac{AR}{AD} = \frac{AP}{AC}$$
$$\frac{AQ}{AB} = \frac{AP}{AC}$$
$$\frac{AR}{AD} = \frac{AQ}{AB}$$

$\frac{AR}{AD} = \frac{AQ}{AB}$ (Proved)



#LP: In the fig, if $PQ \parallel BC$ and $PR \parallel CD$. Prove that

i. $\frac{AQ}{AB} = \frac{AR}{AC}$ ii. $\frac{QB}{AQ} = \frac{DR}{AR}$

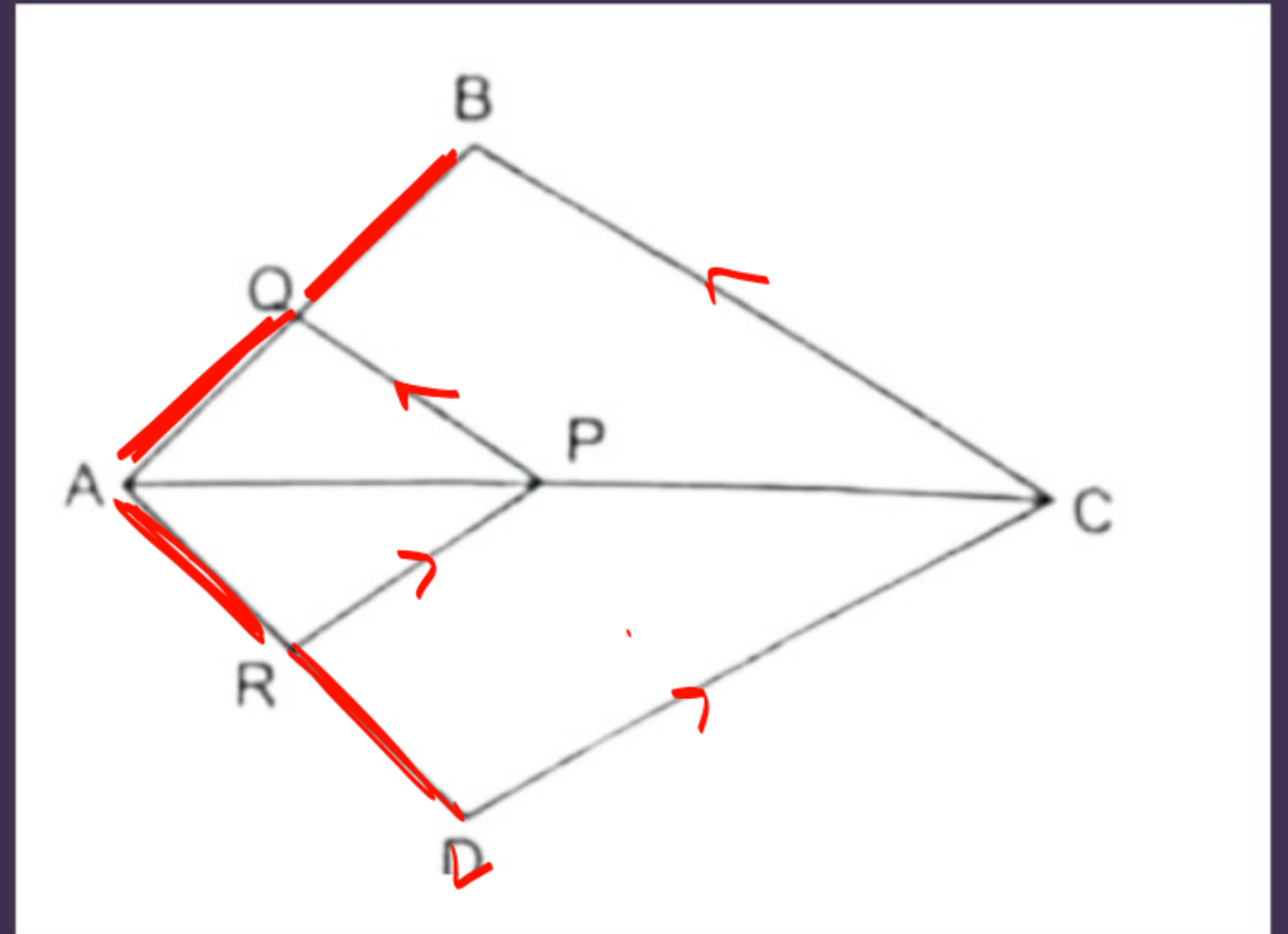
[CBSE 2010]

$$\frac{AR}{RD} = \frac{AP}{PC} \quad \frac{AQ}{QB} = \frac{AP}{PC}$$

$$\frac{AR}{RD} = \frac{AQ}{QB}$$

Reciprocal

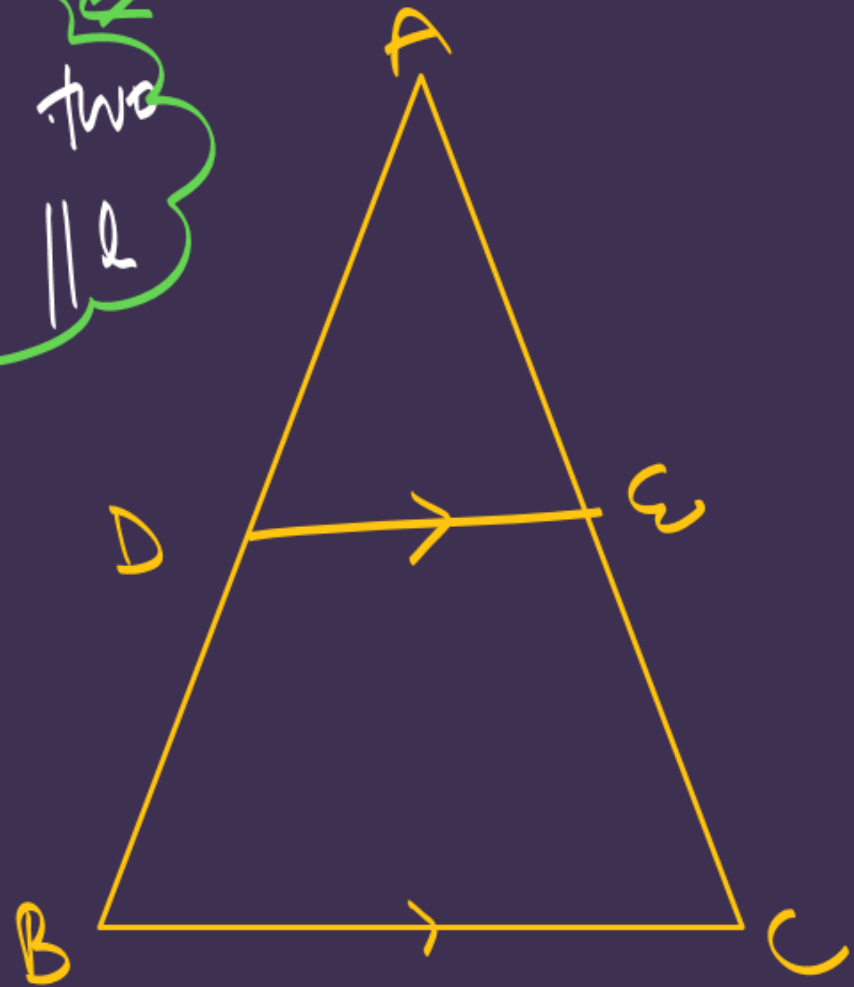
$$\frac{RD}{AR} = \frac{QB}{AQ}$$



#K3D

BPT

given $\frac{AD}{DB} = \frac{AE}{EC}$ two lines $\parallel$



if $DE \parallel BC$

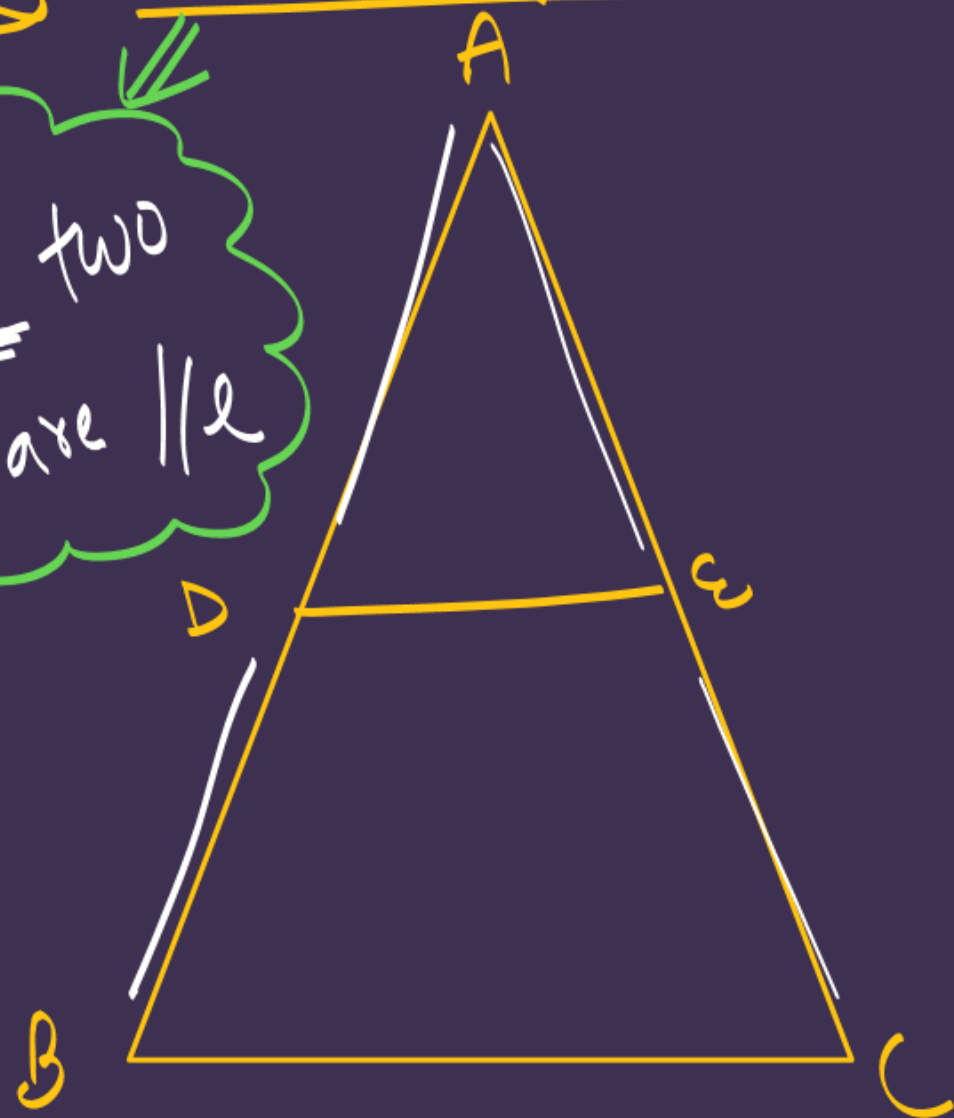
then,

$$\frac{AD}{DB} = \frac{AE}{EC}$$

Converse

Converse of BPT

to prove two lines are $\parallel$



$$\text{if } \frac{AD}{DB} = \frac{AE}{EC}$$

then, $DE \parallel BC$

#CONVERSE OF BPT:

If a line divides any two sides of a triangle in the same ratio, then the line must be
parallel to the third side.

#LP: In the fig, A, B and C are points on OP, OQ and OR respectively such that $AB \parallel PQ$ and $BC \parallel QR$. Show that $AC \parallel PR$.

$\triangle OPQ$

$\text{BPT} \Rightarrow \frac{OB}{BQ} = \frac{OA}{AP}$



$\text{BPT } (\triangle OQR)$

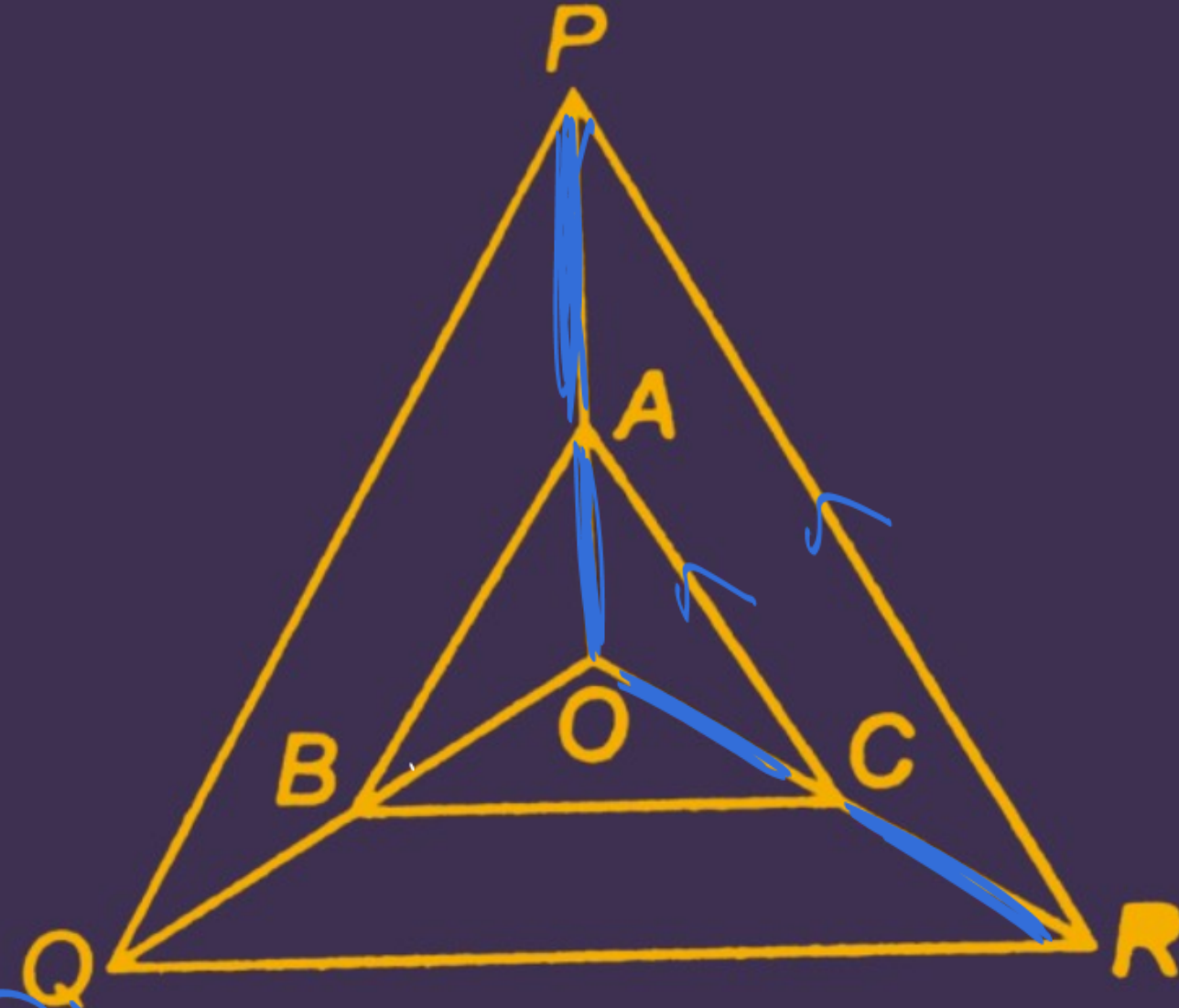
$$\frac{OB}{BQ} = \frac{OC}{CR}$$

from (I) & (II)

$$\frac{OA}{AP} = \frac{OC}{CR}$$

by converse
of BPT

$$\boxed{AC \parallel PR}$$



~~#K3B~~

// given है $\rightarrow$ BPT

// prove करना है $\rightarrow$ Converse of BPT

#LP: In the fig, if $DE \parallel AQ$ and $DF \parallel AR$. Prove that $EF \parallel QR$.

[CBSE 2008]

($\triangle PQA$) BPT

$$\frac{PE}{EQ} = \frac{PD}{DA} \quad (1)$$

by BPT ($\triangle PAR$)

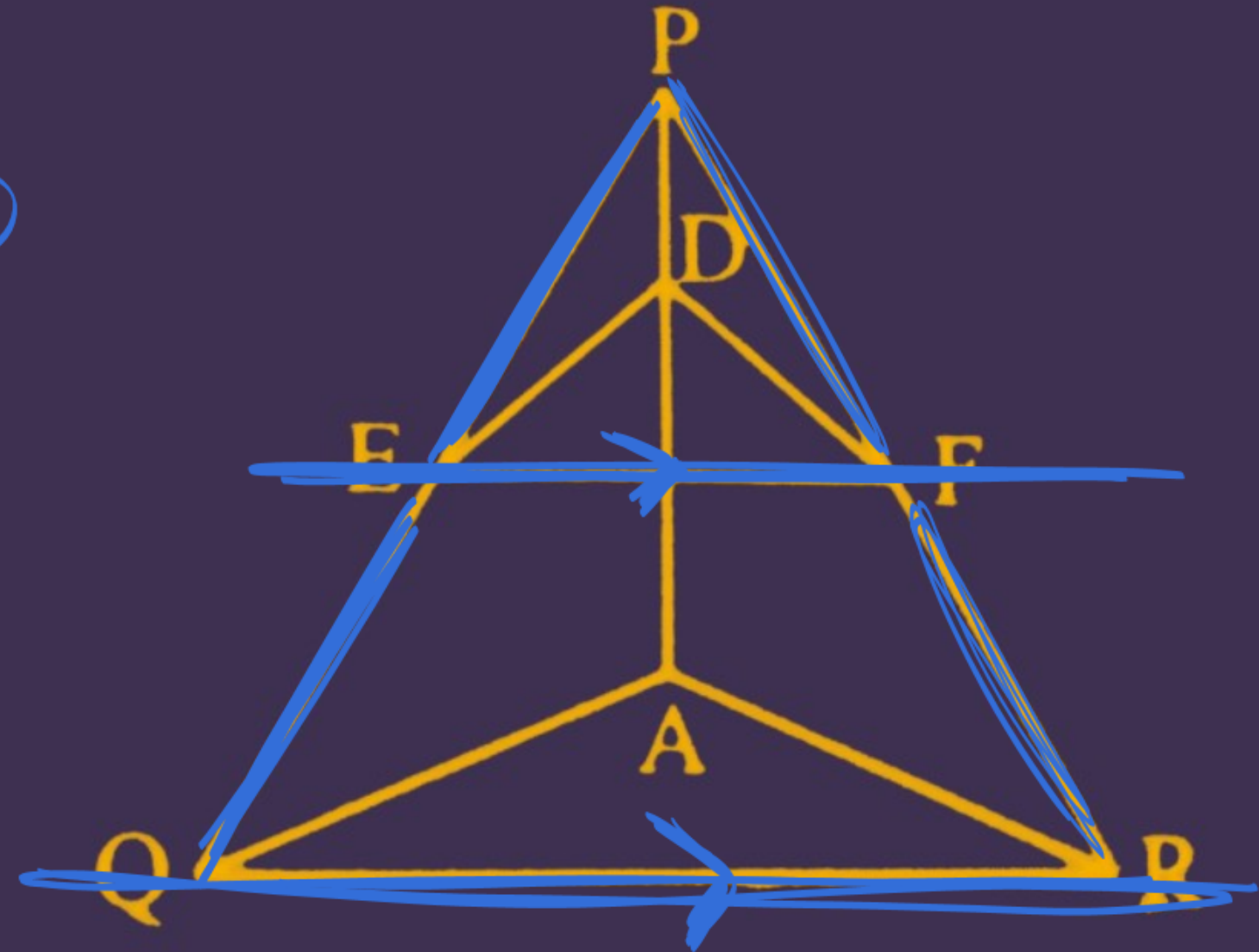
$$\frac{PD}{DA} = \frac{PF}{FR} \quad (2)$$

from (1) & (2)

$$\frac{PE}{EQ} = \frac{PF}{FR}$$

by converse
of BPT $\Rightarrow$

$$EF \parallel QR \quad (3)$$



Thank you Goodieess !!

